Satellite Orbital Collision Risk Prediction

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1 Abstract

We developed a machine learning pipeline to predict orbital collision risk using real conjunction event data from the Kaggle Collision Avoidance Challenge (162,634 labeled records, 13,154 unique events). The continuous risk score was discretized into three classes (Low/Medium/High) via training-set quantile thresholds. After preprocessing (median imputation of 50 columns, 4 missingness flags, log-transforms, RobustScaler) and domain-driven feature engineering (37 new features spanning relative geometry, orbital differences, uncertainty propagation, observation quality, and space weather interactions), four models were compared: Logistic Regression, Random Forest, XGBoost, and a Deep Neural Network. XGBoost achieved 89.63% accuracy and 0.8809 Macro F1-score, followed by the DNN at 0.8427 Macro F1.

2 Introduction

Earth's orbital environment is increasingly congested due to satellite constellation expansion (Starlink, OneWeb), commercial launches, and debris accumulation. Even 1 cm fragments at orbital velocities (78 km/s) carry kinetic energy comparable to a hand grenade. Conjunction analysis and collision avoidance are now critical components of space operations managed by the 18th Space Defense Squadron and ESA's Space Debris Office.

Traditional methods rely on covariance propagation and manually interpreted probability-of-collision thresholdscomputationally expensive and sensitive to atmospheric drag uncertainty. Machine learning can learn complex nonlinear relationships from historical conjunction data for faster, automated risk prioritization at scale.

Objectives:

(1) Formulate risk prediction as multiclass classification for operational decisions

(2) build a preprocessing and feature engineering pipeline respecting group-based data boundaries

(3) compare classical ML models against a deep neural network

(4) identify which paradigms are most effective for structured orbital data.

3 Dataset Description

Each record is a Conjunction Data Message (CDM) capturing physical, geometric, observational, and uncertainty-related measurements at a specific time before the Time of Closest Approach (TCA). The 103 columns comprise 98 (float64), 4 (int64), and 1 (categorical).

Each event_id appears ∼12.4 times on average (max 23), representing multiple CDMs for the same conjunction. GroupShuffleSplit guaranteed zero event overlap, preventing leakage.

Partition	Records	Events	Purpose
Full labeled set	162,634	13,154	Model development
Training split	130,596	∼10,523	Training
Validation split	32,038	∼2,631	Evaluation
Unlabeled test	24,484	Separate	Final predictions

Table 1: Dataset partitions using GroupShuffleSplit on event_id (seed=42) with zero event overlap

4 Problem Formulation

The continuous risk score (log-scaled collision probability, range $[-30.0, -1.44]$, mean -19.34, std 10.01) was discretized using training-set quantile thresholds. 41.3% of records (67,240 rows) were floored at -30.0; only 28.2% (45,793 rows) exceeded -10.
Validation proportions matched closely (49.88/30.08/20.03%). The 50/30/20 imbalance was addressed via class-weighted loss functions in all models.

Class	Label	Threshold	Train Count	Train %
0	Low Risk	risk ≤ -17.871 (50th pctl)	65,299	50.00%
1	Medium Risk	$-17.871 <$ risk ≤ -8.098 (80th)	39,179	30.00%
2	High Risk	risk > -8.098	26,118	20.00%

Table 2: Target label construction from training-set quantiles

5 Data Preprocessing

5.1 Missing Values

50 of 100 feature columns had missing values (31 columns > 5%, 5 columns 15%, 14 columns < 1%):

Column Group (Cols)	Missing %	Physical Interpretation
c_rcs_estimate (1)	32.49%	Debris objects lack radar cross-section
Target covariance cross-terms (12)	5.68% ea.	Incomplete target orbit determination
Chaser cov. cross-terms + σ_{dot} (18)	5.68% ea.	Incomplete chaser orbit determination
Solar indices: F10, F3M, SSN, AP (4)	4.19% ea.	Space weather data collection gaps
t_rcs_estimate (1)	2.01%	Target RCS unavailable
Chaser observation columns (14)	< 0.01% ea.	Rare chaser tracking data gaps

Table 3: Missing value summary by group

5.2 Pipeline

Applied sequentially, all fitted on training fold only:

1. Missingness flags: 4 binary columns (flag_c_rcs_missing, flag_t_rcs_missing, flag_cov_block_missing, flag_solar_missing) created before imputation.

2. One-hot encoding: c_object_type → 5 binary columns. Count: 100 → 108.

3. Median imputation: All 50 missing columns. Median chosen over mean due to extreme skewness (c_weighted_rms: skew = 112.5, t_time_lastob_start: skew = 80.6).
4. Log1p transform: $\log(1 + x)$ on right-skewed non-negative columns. Binary flags excluded.
5. RobustScaler: Centering on median, scaling by IQR. Applied to 99 numeric columns.

6 Feature Engineering

37 domain features were engineered across 6 categories (plus 4 flags + 5 one-hot = 46 new columns), increasing the feature space from 100 raw to 145 columns.
Leakage prevention was applied by removing three target-proxy variables (max_risk_estimate, max_risk_scaling, and risk_estimate_gap), resulting in a final feature set of 142 predictors. For numerical stability, the feature cov_det_ratio was clipped to the range $[-10^6, 10^6]$, since its original maximum value reached 3.42×10^{50}, far exceeding standard float32 limits.

Feature	**Formula**	**Rationale**
rel_pos_mag	$\sqrt{r^2 + t^2 + n^2}$	Euclidean distance in RTN frame
rel_vel_mag	$\sqrt{v_r^2 + v_t^2 + v_n^2}$	Total approach speed
miss_per_speed	miss_dist / rel_speed	Time-like impact proxy
pos_r_over_t	pos_r / pos_t	Radial vs. along-track ratio

Table 4: Relative geometry features (RTN frame)

Feature	**Formula**	**Rationale**
delta_sma/ecc/inc (signed + abs)	t − c	Orbit altitude, shape, and plane differences
delta_h_apo, delta_h_per	t_h − c_h	Apogee/perigee height differences
ecc_product	t_ecc × c_ecc	Joint eccentricity interaction
combined_rcs, rcs_ratio	sum, ratio	Total and relative object size (RCS)
combined_cd/cr_area	sum	Drag and reflectivity area-to-mass

Table 5: Orbital parameter differences and cross-section features

Feature	**Formula**	**Rationale**
combined_sigma_r/t/n	$\sqrt{\sigma_t^2 + \sigma_c^2}$	Joint positional uncertainty (RSS)
total_pos_uncertainty	$\sqrt{\sigma_r^2 + \sigma_t^2 + \sigma_n^2}$	Total 3D positional uncertainty
miss_over_sigma	miss / total_pos_unc	**Top predictor.** Uncertainty-normalized miss distance
cov_det_ratio	t_det / c_det	Ratio of covariance determinants
t/c_obs_usage_ratio	used / available	Fraction of observations used in OD
t_od_span_ratio	actual / recommended	OD window coverage
sigma_r_per_tca, miss_per_tca	feature / time_to_tca	Uncertainty and miss distance evolution rates
solar_activity, solar_pressure	F10×SSN, F10×AP	Space weather interaction terms

Table 6: Uncertainty, observation quality, and interaction features

7 Models and Methodology

All models trained on 130,596 rows, evaluated on 32,038 rows.

Logistic Regression (Baseline): Multinomial softmax, lbfgs solver, max_iter=2000, class_weight=balanced. Tests whether linear decision boundaries suffice for the 142-dimensional feature space.

Random Forest: Bagged ensemble of decision trees learning nonlinear interactions.

Parameter	Value	Rationale
n_estimators	300	Sufficient for variance reduction
max_depth	12	Limits overfitting
min_samples_split	5	Prevents small-node splits
min_samples_leaf	2	Minimum leaf support
class_weight	balanced	Adjusts for 50/30/20 distribution

Table 7: Random Forest hyperparameters

XGBoost: Gradient-boosted trees; sequential boosting corrects residual errors.

Parameter	Value	Rationale
n_estimators	300	Boosting rounds
max_depth	6	Shallower (boosting compensates)
learning_rate	0.05	Gradual error correction
subsample	0.9	Stochastic gradient boosting
colsample_bytree	0.9	Feature sampling per tree
eval_metric	mlogloss	Multiclass log-loss

Table 8: XGBoost hyperparameters

Deep Neural Network (PyTorch): Architecture in Figure 1.

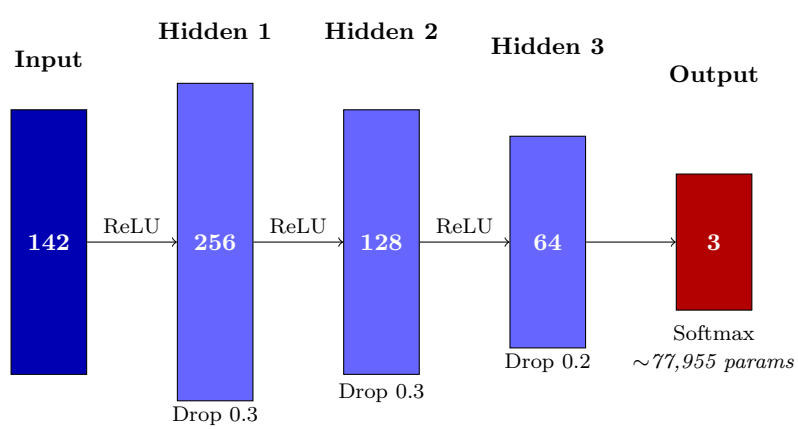

Figure 1: DNN architecture. Layer heights represent relative dimensionality.

Training: Adam, lr=0.0005 (reduced from 0.001 which oscillated); CrossEntropyLoss with weights [0.667, 1.111, 1.667]; batch=256; 20 epochs (best at epoch 14); CUDA. StandardScaler applied on top of RobustScaler for DNN only.

8 Evaluation Metrics

Macro F1-score was the primary metric because it treats all three classes equally regardless of support, making it sensitive to minority-class failures that accuracy would mask (predicting only Low Risk yields 50% accuracy). Accuracy, Macro Precision, and Macro Recall were also reported.

9 Experimental Results

9.1 Summary Comparison

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
XGBoost	**0.8963**	**0.8798**	**0.8823**	**0.8809**
DNN	0.8619	0.8362	0.8514	0.8427
Random Forest	0.7903	0.7560	0.7779	0.7631
Logistic Regression	0.4988	0.1663	0.3333	0.2219

Table 9: Validation performance comparison

9.2 Per-Class Classification Reports

	Logistic Reg.			**Random Forest**			**XGBoost**			**DNN**		
Class	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Low	.50	1.00	.67	.95	.85	.90	.96	.94	**.95**	.97	.90	.93
Medium	.00	.00	.00	.68	.68	.68	.82	.85	**.84**	.77	.80	.78
High	.00	.00	.00	.64	.80	.71	.86	.85	**.86**	.77	.85	.81

Table 10: Per-class results for all models (Support: Low = 15,982; Medium = 9,638; High = 6,418)

Logistic Regression collapsed to predicting all samples as Low Risk (100% recall for Class 0, zero for others), confirming nonlinear class boundaries. **Random Forest** achieved strong Low Risk F1 (0.90) but confused Medium/High (F1: 0.68/0.71), over-predicting High Risk. **XGBoost** achieved the best balance with all per-class F1 > 0.84. The 16-point F1 gain on Medium (0.84 vs. 0.68) is the largest single-class improvement. **DNN** had the highest Low Risk precision (0.97) but underperformed XGBoost on Medium/High by 4–6 F1 points.

9.3 DNN Training Dynamics

The Deep Neural Network was trained for 20 epochs using the Adam optimizer with a learning rate of 0.0005 and class weighted cross entropy loss to address class imbalance. Validation Macro F1 improved consistently during training, increasing from 0.6965 in the first epoch to a peak of 0.8473 at Epoch 18, before stabilizing in later epochs. This trend indicates that the network successfully learned meaningful nonlinear relationships

from the engineered features while avoiding severe overfitting. The best checkpoint was selected based on validation Macro F1, showing that moderate training duration with proper scaling and regularization was effective for this tabular classification task.

10 Feature Importance

Both Random Forest (Gini importance) and XGBoost (gain importance) identified similar variables as the most influential predictors:

1. miss_over_sigma: Ranked as the top feature in both models. This variable represents miss distance normalized by total positional uncertainty, capturing the key conjunction principle that risk depends on proximity relative to uncertainty.

2. mahalanobis_distance and total_pos_uncertainty: These features ranked highly in both models, indicating that covariance structure and combined positional uncertainty are critical for collision risk estimation.

3. Observation quality features (obs_usage_ratio, od_span_ratio): Appeared within the top-ranked variables, suggesting that tracking reliability and orbit determination quality are themselves predictive of conjunction risk.

4. Time-to-TCA features (time_to_tca, sigma_r_per_tca): Important temporal variables that capture how uncertainty evolves as the event approaches the time of closest approach.

Raw geometric features such as miss distance alone or position components ranked below their uncertainty-normalized counterparts, confirming that domain-engineered features grounded in orbital mechanics provided stronger predictive signal than raw measurements alone.

11 Discussion

XGBoost dominated because gradient boosting captures complex nonlinear interactions by sequentially correcting residuals across 300 depth-6 trees. Collision risk depends on entangled geometry–uncertainty–observation relationships that boosting handles well.

DNN became competitive after StandardScaler standardization but underperformed XGBoost, consistent with Grinsztajn et al. (NeurIPS 2022). Likely causes: (1) FC architectures don't exploit tree-like splits natural to tabular data; (2) 130K samples insufficient for 142 inputs without stronger priors.

Random Forest's 12-point Macro F1 gap (0.763 vs. 0.881) reflects bagging's inability to correct systematic errors. **Logistic Regression's** collapse confirms nonlinear class boundaries.

Operational implications: XGBoost's per-class F1 scores are operationally significant:

- **High Risk recall of 0.85:** 85% of dangerous encounters are correctly flagged. The 15% miss rate must be evaluated against the cost of missed detections.

- **High Risk precision of 0.86:** 86% of flagged encounters are genuinely high-risk. The 14% false-alarm rate is acceptable for a first-pass screening tool.

- **Medium Risk F1 of 0.84:** The hardest boundary still performs well, reducing analyst workload for ambiguous cases.

The model supplements, not replaces, physics-based P_c calculations.

12 Reproducibility and Conclusion

Notebooks:

- 01_EDA_preprocessing.ipynb: Data loading, 50-column missing analysis, imputation, one-hot encoding, scaling, label generation.
- 02_Feature_Engineering.ipynb: 37 domain features, leakage prevention, cov_det_ratio clipping.
- 03_Modeling_and_Comparison.ipynb: Centralized CONFIG, 4 models, per-class reports, feature importance.

Reproducibility:
Reproducibility was ensured through a consistent experimental setup. A fixed random seed (seed = 42) was applied across all models, data splits, and PyTorch initialization.
Training and validation sets were generated using GroupShuffleSplit(random_state = 42). All preprocessing components, including quantile thresholds ($q_{50} = -17.871$, $q_{80} = -8.098$), imputers, and scalers, were fitted exclusively on the training fold to prevent data leakage.
A centralized CONFIG dictionary was used to manage experiment parameters, and identical evaluation functions were applied across all models to guarantee fair and consistent comparison.

GitHub Repo Link: `https://github.com/siddhi07/Satellite-Orbital-Collision-Risk-Prediction`

Conclusion:
This project built a complete ML pipeline for satellite collision risk prediction using 162,634 conjunction records across 13,154 events. Key contributions:

- Rigorous group-based splitting preventing event-level leakage.
- 37 domain features encoding orbital mechanics principles.
- Systematic four-model comparison.
- Identification of miss_over_sigma as the dominant predictor.

XGBoost achieved **0.8809 Macro F1** and **89.63% accuracy**, confirming gradient-boosted trees as optimal for structured tabular orbital data.

Future work:

- Time-series CDM sequence modeling.
- Graph-based orbital interaction networks.
- Atmospheric density integration.
- Real-time operational deployment.

Glossary of Terms and Definitions

Space Operations & Orbital Mechanics

CDM — Conjunction Data Message; standardized close-approach data from space surveillance organizations containing orbital states, covariances, and metadata.

TCA — Time of Closest Approach; the predicted moment when two orbiting objects reach minimum separation distance.

Conjunction — An event where two objects are predicted to pass within a specified distance threshold, triggering collision avoidance analysis.

Miss Distance — Predicted minimum geometric separation at TCA, measured in kilometers.

P_c — Collision probability computed by integrating probability density over combined cross-sectional area. The dataset's risk score is $\log P_c$.

Maneuver — Deliberate orbit change (using thrusters) to avoid a predicted collision; costly in fuel and mission time.

RTN Frame — Radial-Transverse-Normal local orbital coordinate system. R = radial from Earth center, T = along velocity, N = orbit normal.

SMA — Semi-major axis; half the longest orbit diameter, determines altitude and period.

Eccentricity — Orbit shape parameter (0 = circular, approaching 1 = highly elliptical).

Inclination — Angle between the orbital plane and Earth's equatorial plane, in degrees.

Apogee/Perigee — Farthest/nearest orbital points from Earth's surface, in km.

Uncertainty & Tracking

Covariance Matrix — 6×6 symmetric matrix encoding position and velocity uncertainties and their cross-correlations. Determinant measures uncertainty ellipsoid volume.

Sigma (σ) — Standard deviation of position or velocity estimate in a given direction (e.g., σ_r = radial position uncertainty).

Mahalanobis Dist. — Multi-dimensional distance metric accounting for correlations; measures uncertainty-scaled separation between objects.

OD — Orbit Determination; estimating orbital parameters from tracking observations (radar, optical, laser).

RCS — Radar Cross Section; measure of radar detectability correlated with physical size (m^2). 32.49% missing for chaser objects.

F10.7 — Solar radio emission at 10.7 cm; proxy for solar activity driving atmospheric density and LEO satellite drag.

SSN — Sunspot number; correlates with solar activity.

AP — Geomagnetic disturbance index; high values indicate storms expanding the atmosphere.

Machine Learning

Macro F1 — Unweighted mean of per-class F1-scores: $(F1_0+F1_1+F1_2)/3$. Sensitive to minority-class performance.

GroupShuffleSplit — Splitter keeping all samples with the same group key (event_id) in one fold; prevents data leakage.

XGBoost — Gradient-boosted tree ensemble; sequentially trains trees correcting residuals with regularization.

RobustScaler — Scales features by subtracting median and dividing by IQR; resistant to outliers.

Dropout — Regularization technique randomly zeroing neuron activations during training to prevent co-adaptation.

Data Leakage — When out-of-training information influences model training, artificially inflating performance metrics.

Object Types

DEBRIS — Non-functional man-made objects in orbit (58.1% of dataset).

PAYLOAD — Active or inactive satellites (9.5%).

ROCKET BODY — Spent upper stages (1.2%).

UNKNOWN/TBA — Unclassified objects (31.2%).

References

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